Department of Mathematics, IIT Madras
MA2101 Linear Algebra via Matrices
Assignment 3

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1. Let U be a subspace of V and let V be a subspace of W, where W is a subspace of $\mathbb{F}^n$. Is U a subspace of W?
2. Let $u, v_1, v_2, \ldots, v_n$ be $n+1$ distinct vectors in $\mathbb{F}^n$, $S_1 = \{v_1, v_2, \ldots, v_n\}$, and let $S_2 = \{u, v_1, v_2, \ldots, v_n\}$. Prove that $\text{span}(S_1) = \text{span}(S_2)$ iff $u \in \text{span}(S_1)$.
3. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove the following:
   (a) A is a subspace of $\mathbb{F}^n$ if and only if $\text{span}(A) = A$.
   (b) If $A \subseteq B$, then $\text{span}(A) \subseteq \text{span}(B)$.
   (c) $\text{span}(A \cup B) = \{u + v : u \in \text{span}(A),\ v \in \text{span}(B)\}$.
   (d) $\text{span}(A \cap B) \subseteq \text{span}(A) \cap \text{span}(B)$.
   (e) $\text{span}(A) \cap \text{span}(B) \subseteq \text{span}(A \cap B)$.
4. Let $S \subseteq U \subseteq \mathbb{F}^n$, where U is a subspace of $\mathbb{F}^n$. Suppose for any subspace V of $\mathbb{F}^n$, if $S \subseteq V$ then $U \subseteq V$. Prove that $U = \text{span}(S)$.
5. Let A and B be subsets of $\mathbb{F}^n$. Prove or disprove: $A \cup B$ is linearly independent iff $\text{span}(A) \cap \text{span}(B) = \{0\}$.
6. Let V be a subspace of $\mathbb{F}^n$ and let $B \subseteq V$. Prove the following:
   (a) B is a basis of V iff B is linearly independent and each proper superset of B is linearly dependent.
   (b) B is a basis of V iff B spans V and no proper subset of B spans V.
7. Let $\{x, y, z\}$ be a basis for a vector space V. Is $\{x + y, y + z, z + x\}$ also a basis for V?
8. Find a basis for the subspace $\{(a, b, c) \in \mathbb{R}^3 : a + b - 5c = 0\}$ of $\mathbb{R}^3$.
9. Find bases and dimensions of the following subspaces of $\mathbb{R}^5$:
   (a) $\{(a, b, c, d, e) \in \mathbb{R}^5 : a - c - d = 0\}$.
   (b) $\{(a, b, c, d, e) \in \mathbb{R}^5 : b = c = d,\ a + e = 0\}$.
   (c) $\text{span}\{(1, -1, 0, 2, 1),\ (2, 1, -2, 0, 0),\ (0, -3, 2, 4, 2),$
   $(3, 3, -4, -2, -1),\ (5, 7, -3, -2, 0)\}$.
10. Extend the set $\{(1, 0, 1, 0),\ (1, 0, -1, 0)\}$ to a basis of $\mathbb{R}^4$.
11. Let $u_1, v, w, x_1, x_2, x_3, x_4 \in \mathbb{R}^5$ satisfy $x_1 = u + 2v + 3w$, $x_2 = 2u - 3v + 4w$, $x_3 = -3u + 4v - 5w$ and $x_4 = -u + 6v + w$. Is $\{x_1, x_2, x_3, x_4\}$ linearly dependent?
12. Prove that the only proper subspaces of $\mathbb{R}^2$ are the straight lines passing through the origin.
13. Describe all subspaces of $\mathbb{R}^3$.
14. Let $A \in \mathbb{F}^{n \times n}$. Let $\{v_1, \ldots, v_n\}$ be any basis of $\mathbb{F}^{n \times 1}$. Prove that A is invertible iff $Ax = v_i$ has at least one solution for each $i = 1, \ldots, n$.
15. Let $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 \\ 3 & 5 & 3 & 4 & 3 \\ 1 & 1 & 1 & 2 & 1 \\ 5 & 8 & 5 & 7 & 5 \end{bmatrix}$. Write $r = \text{rank}(A)$ and $k = \text{null}(A)$.
   (a) Find bases for $R(A)$ and $N(A)$, and then determine r and k.
   (b) Express suitable $4 - r$ rows of A as linear combinations of other r rows.
   (c) Express suitable $5 - r$ columns of A as linear combinations of other r columns.
16. Construct bases for (i) the row space, (ii) the column space, and the null space for each of the following matrices:

(a) $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 1 & 7 \\ 2 & 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & -3 \\ 1 & 2 & 8 \\ 3 & -1 & 4 \\ 4 & -2 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 2 \\ 1 & 3 & -2 & 1 \end{bmatrix}$

17. Let $A \in \mathbb{F}^{n\times n}$. Consider A as the linear transformation $A : \mathbb{F}^{n\times 1} \to \mathbb{F}^{n\times 1}$ given by $A(x) = Ax$ for each $x \in \mathbb{F}^{n\times 1}$. Prove the following:
    (a) A is one-one iff $N(A) = \{0\}$.
    (b) A is one-one and onto iff A maps any basis onto a basis of $\mathbb{F}^{n\times 1}$.
    (c) A is one-one iff A is onto.
18. Let A and B be matrices having the same RREF. Then which of the following may be concluded? Explain.
    (a) $R(A) = R(B)$ (b) $N(A) = N(B)$
    (c) The row space of A is equal to the row space of B.
19. For $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$, prove: $\text{rank}(AB) \le \min\{\text{rank}(A), \text{rank}(B)\}$.
20. In the following a basis B for $\mathbb{F}^{3\times 1}$ and a vector $u \in \mathbb{F}^{3\times 1}$ are given. Compute the coordinate vector $[u]_B$.
    (a) $B = \{[0,0,1]^t,\ [1,0,0]^t,\ [0,1,0]^t\},\ u = [1,2,3]^t$
    (b) $B = \{[0,1,1]^t,\ [1,0,1]^t,\ [1,1,0]^t\},\ u = [2,1,3]^t$
    (c) $B = \{[1,1,1]^t,\ [1,2,1]^t,\ [1,2,3]^t\},\ u = [3,2,1]^t$
21. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$. Consider the bases $B = \{[1,0,1]^t,\ [0,1,1]^t,\ [1,1,0]^t$ and $C = \{[1,1,1]^t,\ [1,2,1]^t, [1,1,0]^t\}$ for $\mathbb{F}^{3\times 1}$. Compute $[A]_{C,B}$ and $[A]_{B,C}$ in the following ways:
    (a) by expressing A-images of the basis vectors of B in terms of those of C and vice versa.
    (b) by computing the inverses of two matrices constructed from the basis vectors.
    (c) by using the RREF conversion as mentioned in the text.
22. Let $A, B \in \mathbb{F}^{m\times n}$, C be a basis of $\mathbb{F}^{n\times 1}$, D be a basis for $\mathbb{F}^{m\times 1}$, and let $\alpha \in \mathbb{F}$. Show that $[\alpha A + B]_{D,C} = \alpha[A]_{D,C} + [B]_{D,C}$.
23. Construct a matrix $A \in \mathbb{R}^{2\times 2}$, a vector $v \in \mathbb{R}^{2\times 1}$, and a basis $B = \{u_1, u_2\}$ for $\mathbb{R}^{2\times 1}$ satisfying $[Av]_B \ne A[v]_B$.
24. Let $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 1 & 3 \\ -1 & -1 & -2 \end{bmatrix}$, $V = \{[a,b,c]^t : a+2b+3c = 0,\ a,b,c \in \mathbb{R}\}$, $B = \{[0,3,-2]^t,\ [2,-1,0]^t\}$, and let $C = \{[1,1,-1]^t,\ [3,0,-1]^t\}$.
    (a) Show that V is a 2-dimensional subspace of $\mathbb{R}^{3\times 1}$.
    (b) Show that $A : V \to V$ defined by $A(v) = Av$ for $v \in V$ is a well defined function.
    (c) Show that B and C are bases for V.
    (d) Extend the basis B of V to a basis O for $\mathbb{F}^{3\times 1}$.
    (e) Extend the basis C of V to a basis N for $\mathbb{F}^{3\times 1}$.
    (f) Find the change of basis matrix $I_{N,O}$.
    (g) Verify that $[v]_N = I_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.
    (h) Find the matrix $[A]_{N,O}$.
    (i) Verify that $[Av]_N = [A]_{N,O}\,[v]_O$ for $v = [-4,-1,2]^t$.
25. In each of the following cases, show that T is a linear transformation. Find the matrix A so that $T(x) = Ax$. Determine $\text{rank}(A)$. And then construct the full rank factorization of A.
    (a) $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{2\times 1},\ T([a,b,c]^t) = [c, b+a]^t$
    (b) Let $T : \mathbb{R}^{3\times 1} \to \mathbb{R}^{3\times 1},\ T([a,\ b,\ c]^t) = [a+b,\ 2a-b-c,\ a+b+c]^t$
26. Which matrices are equivalent to, and which are similar to
    (a) the zero matrix (b) the identity matrix?
27. Consider the bases $O = \{u_1, u_2\}$ and $N = \{v_1, v_2\}$ for $\mathbb{F}^{2\times 1}$, where $u_1 = [1,1]^t, u_2 = [-1,1]^t, v_1 = [2,1]^t$, and $v_2 = [1,0]^t$. Let $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) Compute $Q = [A]_{O,O}$ and $R = [A]_{N,N}$.
(b) Find the change of basis matrix $P = I_{N,O}$.
(c) Compute $S = PQP^{-1}$.
(d) Is it true that $R = S$? Why?
(e) If $S = [s_{ij}]$, verify that $Av_1 = s_{11}v_1 + s_{21}v_2,\ Av_2 = s_{12}v_1 + s_{22}v_2$.
